# The Math Behind Printers

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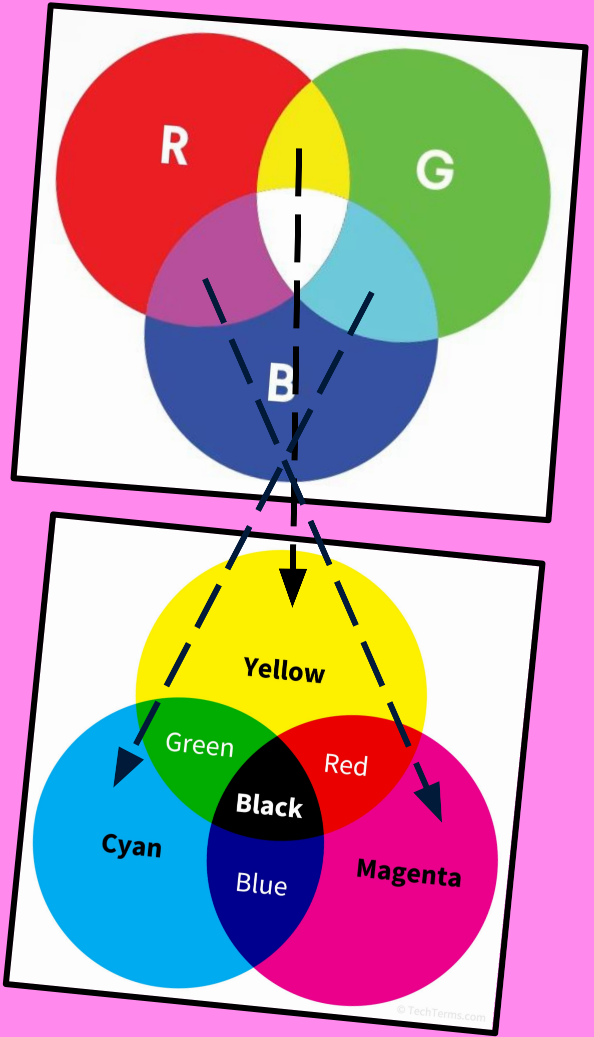
# Color Theory

How do printers work? If their ink was just red, green, and blue, then how could they print black?

Instead, they use a subtractive color model called CMYK. The cyan, magenta, and yellow are the colors you get when you mix red, green, and blue.

- The K in CMYK stands for Key Black.



# RGB → CMYK

## 1. Scale RGB values between 0-1

$R' = R/255$

$G' = G/255$

$B' = B/255$

The R, G, and B values represent an integer 0-255.

## 2. Invert the channels

$C = 1 - R'$

$M = 1 - G'$

$Y = 1 - B'$

We have to subtract because CMYK is a negative model.

# RGB → CMYK

## 3.CMY → CMYK

K is the minimum value of C, M, or Y.

$$K = \min(C, M, Y)$$

## 4. Subtract unnecessary ink

If C, M, and Y are equal, it will print a gray, which wastes ink.

$$C_{new} = (C - K)/(1 - K)$$

$$M_{new} = (M - K)/(1 - K)$$

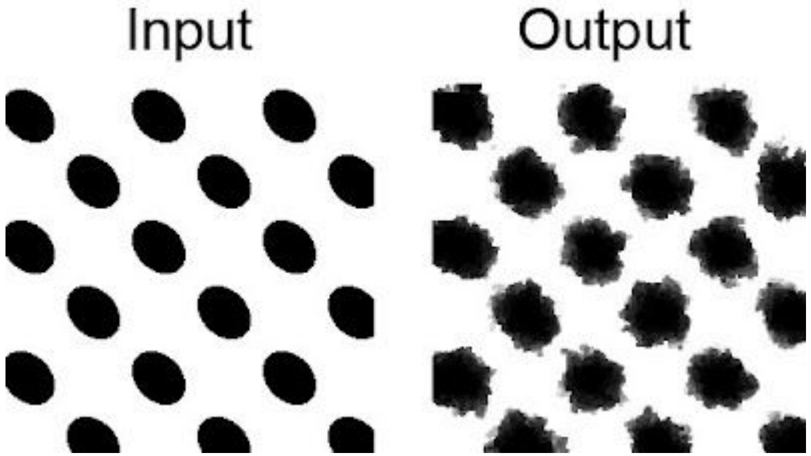
$$Y_{new} = (Y - K)/(1 - K)$$

If K = 1, then C, M, and Y will automatically turn into 0 <u>BEFORE</u> this step.

# The Murray Davies Formula

Gravity, momentum, and capillary suction can result in imperfect dots to be produced when printing.

The Murray Davies Formula provides a mathematical insight on how printers work to prevent this.

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# What causes dots to look bigger?

Paper is made of highly reflective and transluscent fibers. AWhen light enters unprinted white space, it doesn't bounce back. Instead, it bounces around inside the fibers.

So when you look at a dot, the reason it looks bigger than intended is because the light entering the circumference gets trapped under the edge and makes it look larger.

Hello!

# The Light Balance Scale

Printers calculate dot size using light percentages.

R is reflecting light. A is the % of paper covered by ink.

$$R_{total} = a * R_{solid} + (1 - a) * R_{paper}$$

# Optical Density

Using the reflectance we just calculated, we can now see how much light is absorbed.

$$D = -\log_{10}(R)$$

# Calculating Area

Now that we have our density, we can see the area of the circle the printer would make.

$$A = ((1 - 10^{-Dtint})/(1 - 10^{-Dsolid})) * 100$$

This is how printers print utilizing the perfect amount of ink instead of just leaking too much.